The Impact of Overtime Work and Job Dissatisfaction on Employee Attrition

Is employee attrition higher among employees who work overtime compared to those who do not?

Business Question: Do employees that work extra hours leave the company more than who work regular hours?

Why This Matters? Knowing this information will help keep employees for a longer period, and eliminate the costs of hiring and training of the new employees.

Data Description:

Source:” Employee Attrition” - Prashant Patel on Kaggle

Number of records: 1470

Features (used fields):

- Attrition
- Department
- Employee Number
- Environment Satisfaction
- Job Satisfaction
- Overtime
- Relationship Satisfaction.

Tools & Technologies Used:

- Microsoft Excel: Primary data cleaning
- Microsoft Power BI: Searching for trends, visualizations, DAX measures

Data Cleaning & Preparation:

Data set was checked for any duplicates or missing values misspellings. This step is important because clean and prepared data greatly affects its reliability.

Exploratory Data Analysis (EDA):

- The rate of attrition is higher in over-time workers than regular-hour ones**.**
- The research and development department has the most employees.

Techniques used:

- Data visualization; pie charts and bar charts

Data Visualization and Report Creation:

- Clustered column chart: Number of Former Extended-Hour and Regular-Hours Employees
- Pie chart: Percent of Attrition by Department
- Clustered column chart: Former Employees Job Satisfaction Scores
- Clustered column chart: Former Employees Relationship Satisfaction Scores
- Clustered column chart: Former Employees Environmental Satisfaction Scores

DAX measurements:

- What percent of employees in the research and development department work overtime?

  % of Employees in R&D Who Work Overtime =
  CALCULATE(COUNTROWS('Table'), 'Table'[Department] = "Research & Development", 'Table'[OverTime] = "Yes") /
  CALCULATE(COUNTROWS('Table'), 'Table'[Department] = "Research & Development")
- What is the number of former employees?

  Number of Attrition Employee = CALCULATE(COUNTROWS('Table'), 'Table'[Attrition] = "Yes")

Key Insights & findings:

- Working overtime increases job attrition
- The highest percentage of job attrition was in the research & development department
- A significant portion of former employees in the research & development department were dissatisfied with their job, relationships and environment in the workplace.

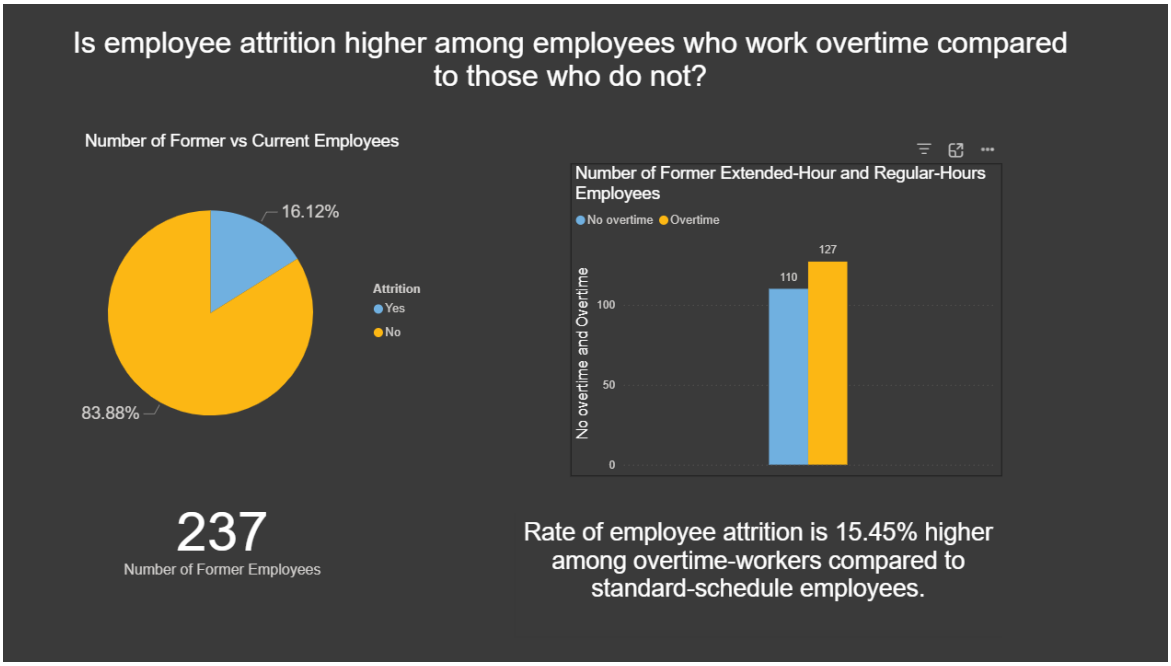
Recommendations:

- Reduce the number of extra hours employees work
- Prioritize creating a healthy workplace environment, strong co-worker relationships and make employees feel appreciated by recognizing their work.

Conclusion:

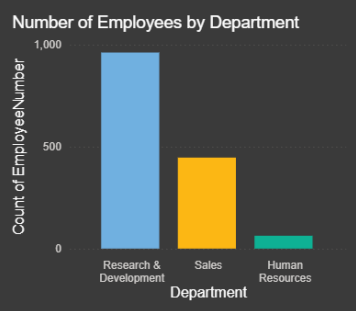
A data set titled Employee Attrition” created Prashant Patel on Kaggle was analyzed for the aim of determining if employees that work overtime leave the company more than standard-hour employees. After analysis, it was found that working overtime and dissatisfaction in job, work environment and relationships in the workplace is associated with high employee attrition. In order to reduce attrition, reduce overtime hours for employees, and create a healthy workplace environment where co-workers have strong relationships, and employees’ work is recognized and appreciated. because frequent changing, hiring and training of new employees is costly.

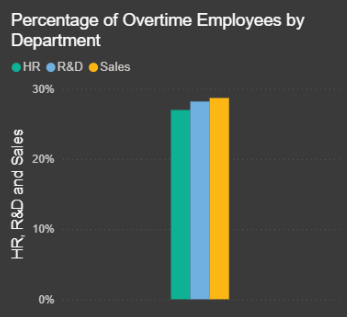
Power BI Report Screenshots:

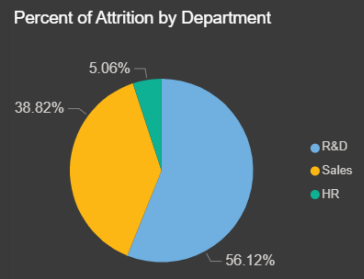


Created By: Alrayyan-A on GitHub

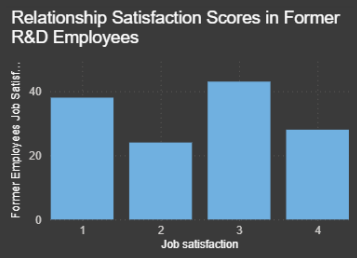
The Research and Development department has the largest workforce. Although the proportion of employees working overtime is relatively consistent across departments, R&D accounts for 56.12% of total employee attrition.







*1= Low, 2= Medium, 3= High, 4= Very High



28.57%
Low Job Satisfaction

28.57%
Low Relationship Satisfaction

32.33%
Low Environment satisfaction

Among former employees in the Research and Development department, 28.57% reported low levels of relationship and overall job satisfaction, while 32.33% experienced low satisfaction with the work environment during their period of employment.